Introduction

Project Name: Tech Sales Performance Dashboard

Dataset: Mock dataset created specifically for this project. Project was generated with a custom Python script.

Purpose: The purpose of this project was to simulate a realistic sales performance dashboard while working with mock data.

Contents

Project Stages:

Preperation

Choosing dataset type and business question. Planning visualised report structure (pages, topics, information which will be presented.)

Mock Data Creation

Mock data generated with Python script (modules used: os, datetime, random, pandas, numpy, faker)

Mock data imported into Power BI, initial data cleaning in Power Query (data type changes.)

Initial data model created, data validtion in Power BI and Excel, adjustment to Python script to make mock data more realistic (market share distribution, weighting order dates, product segments, active customer years.)

Calculations

Measures created and tested either in Power BI or in Dax Studio. Some calculations validated in Excel.

Visualisation

Preparing template design (the current version is a custom design, no themes were used.)

Creating report pages.

Testing

UI testing.

Setting up filtering limitations.

Creating Mobile View

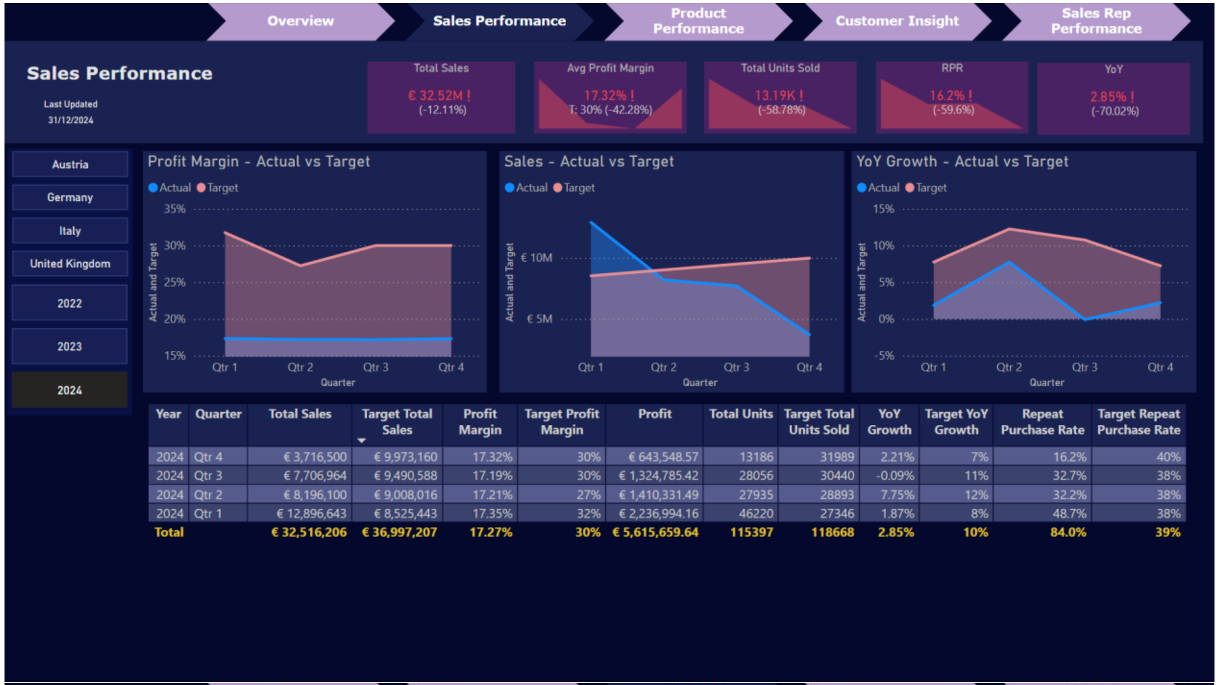
Visualized Report

From my experience, visualisation can be created in many ways depending on business needs. For this project, I have chosen a more general approach and grouped information based on business question subjects (as opposed to, for example, target audience.)

Limitations for this project were mostly in the mock data, but I aimed to generate data that would at least to some degree simulate realistic visuals (although this may not be fully comprehensive.)

Overview

The overview page contains graphs which can give a quick-glance summary of the different parts of the report. In my experience, having a higher-level summary page at the beginning allows senior management to understand the overall performance without having to deep dive into the whole report.



Sales Performance

Business Questions:

How have sales performance evolve throughout the fiscal year?

What was the actual sales performance compared to targets:

What is the comparison between total sales and profit margin.

Product Performance

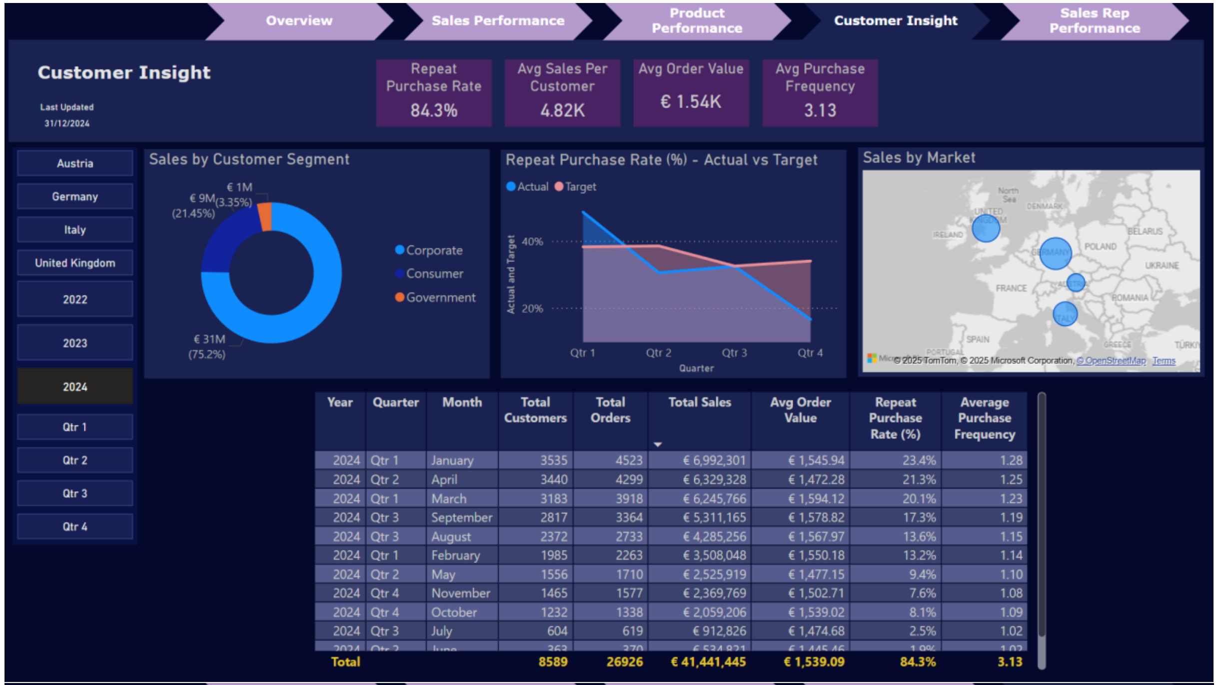
Business Questions

Which products are the most profitable?

What was the sales performance evolution for the different product categories?

Which product sub-category generated the most sales?

How does product discount and unit cost affect the profit margin?



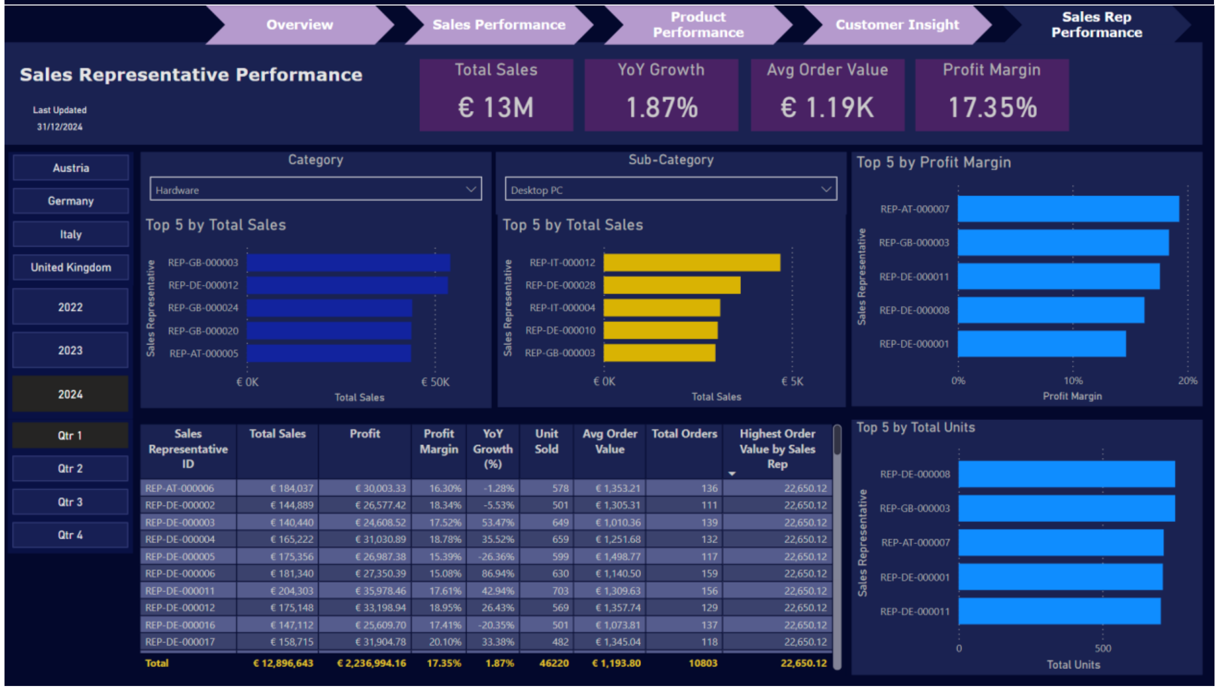
Customer Insight

Business Questions:

How are sales distributed between the different customer segments?

What is the customer distribution between the markets?

What is the customer retention between months and its effect on repeat purchases and purchase frequency?



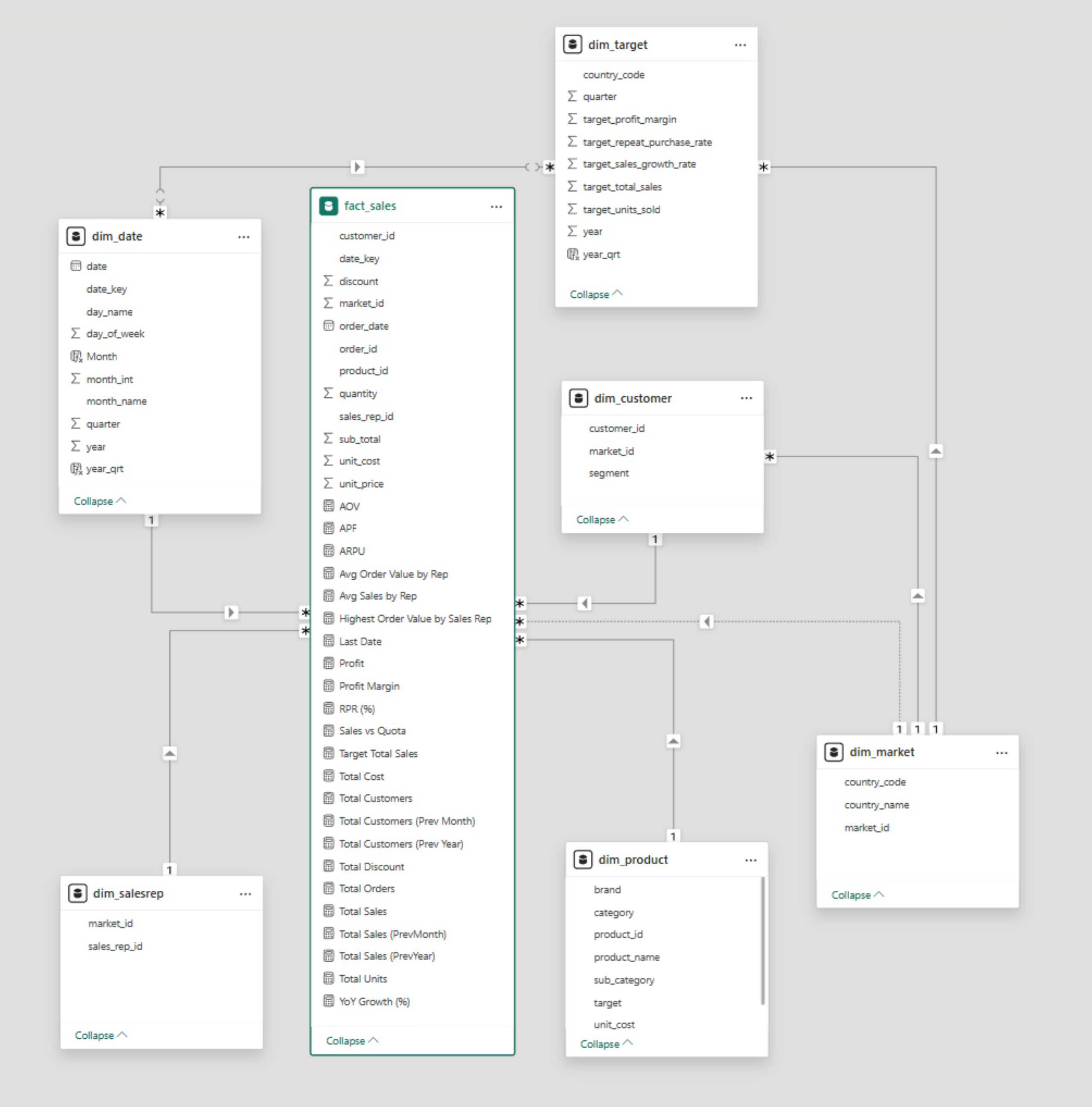
Sales Representative Performance

Business Questions:

Which Sales Representatives generated the most sales for different product categories and sub-categories?

Which Sales Representatives generated the most profit?

Data Model



dim_target
country_code
quarter
target_profit_margin
target_repeat_purchase_rate
target_sales_growth_rate
target_total_sales
target_units_sold
year
year_qrt
Collapse
fact_sales
customer_id
date_key
discount
market_id
order_date
order_id
product_id
quantity
sales_rep_id
sub_total
unit_cost
unit_price
AOV
APF
ARPU
Avg Order Value by Rep
Avg Sales by Rep
Highest Order Value by Sales Rep
Last Date
Profit
Profit Margin
RPR (%)
Sales vs Quota
Target Total Sales
Total Cost
Total Customers
Total Customers (Prev Month)
Total Customers (Prev Year)
Total Discount
Total Orders
Total Sales
Total Sales (PrevMonth)
Total Sales (PrevYear)
Total Units
YoY Growth (%)
Collapse
dim_date
date
date_key
day_name
day_of_week
Month
month_int
month_name
quarter
year
year_qrt
Collapse
dim_customer
customer_id
market_id
segment
Collapse
dim_market
country_code
country_name
market_id
Collapse
dim_product
brand
category
product_id
product_name
sub_category
target
unit_cost
Collapse
dim_salesrep
market_id
sales_rep_id
Collapse

Data Dictionary

Table	Column Name	Description	Data type	Key	Dependencies
dim_market	market_id	unique market identification number	int	primary key	Relationships: dim_customer, dim_salesrep, dim_target, fact_sales
dim_market	country_code	ISO2 market code	str	surrogate key	Relationships: dim_target
dim_market	country_name	market name	str		FILTER
dim_salesrep	sales_rep_id	unique sales representative identification number	int	primary key	Relationships: fact_sales, Measures
dim_salesrep	market_id	market identification number	int	foreign key	Relationships: dim_market
dim_customer	customer_id	unique customer identification number	int	primary key	Relationships: fact_sales
dim_customer	segment	customer segment category	str		FILTER
dim_customer	market_id	unique int id number	int	foreign key	Relationships: dim_market
dim_product	product_id	unique product identification number	int	primary key	Relationships: fact_sales
dim_product	product_name	product name	str		FILTER
dim_product	category	product first level category	str		FILTER
dim_product	sub_category	product second level category	str		FILTER
dim_product	unit_price	product price per unit	float		
dim_product	manufacturer	product manufacturer	int		FILTER
dim_date	date_key	unique date identification number	int	primary key	Relationships: fact_sales
dim_date	date	date (2022-01-01 to 2024-12-31)	date		FILTER
dim_date	year	year of date	int		
dim_date	quarter	quarter of date	int		
dim_date	month_int	month of date	int		
dim_date	month_name	full month string name	str		
dim_date	day_of_week	day of the week as number	int		
dim_date	day_name	full day string name	str		
dim_date	year_qrt	concatenated year and quarter number	int	surrogate key	Relationships: dim_target

fact sales	order_id	unique order identification number	int	primary key	Measures
fact sales	order_date	date the order was completed	date		FILTER, Measures
fact sales	date_key	date of order as date identification number	int	foreign key	Relationships: dim_date
fact sales	customer_id	customer identification number	int	foreign key	Relationships: dim_customer, Measures
fact sales	product_id	product identification number	int	foreign key	Relationships: dim_product
fact sales	sales_rep_id	sales representative identification number	int	foreign key	Relationships: dim_salesrep, Measures
fact sales	market_id	market identification number	int	foreign key	Relationships: dim_market
fact sales	quantity	total quantity of products in the order	int		Measures
fact sales	unit_price	unit price of product in the order	float		
fact sales	unit_cost	cost of the unit	float		Measure
fact sales	discount	discount applied to the order	float		Measure
fact sales	sub_total	sub total of the order	float		Measures
dim_target	country_code	relevant ISO2 market code for the objective	str	surrogate key	Relationships: dim_market
dim_target	year	relevant year for the objective	int		year_qrt field
dim_target	quarter	relevant quarter for the objective	int		year_qrt field
dim_target	target_profit_margin	profit margin objective	float		VISUALISATION
dim_target	target_repeat_purchase_rate	repeat purchase rate objective	float		VISUALISATION
dim_target	target_sales_growth_rate	year over year objective	float		VISUALISATION
dim_target	target_total_sales	total sales objective	float		VISUALISATION
dim_target	target_units_sold	total units sold objective	int		VISUALISATION
dim_target	year_qrt	concatenated year and quarter number	int	surrogate key	Relationships: dim_date

Dax Measures

MEASURE NAME	DESCRIPTION	EXPRESSION	MEASURE DISPLAY FOLDER	DEFAULT FORMAT STRING
Total Sales		SUMX(fact_sales, fact_sales[sub_total])	Measures\Sales Measures	"â,¬"\ #,0;-"â,¬"\ #,0;"â,¬"\ #,0
Total Units		Sum(fact_sales[quantity])	Measures\Sales Measures	0
Total Discount		SUMX(fact_sales, fact_sales[unit_price] * fact_sales[Quantity] * fact_sales[Discount])	Measures\Sales Measures	
Total Cost		SUMX(fact_sales, fact_sales[unit_cost] * fact_sales[quantity])	Measures\Sales Measures	
Sales vs Quota		DIVIDE([Total Sales], AVERAGE(dim_salesrep[Quota]))	Measures\Sales Measures	
Profit		[Total Sales]-[Total Cost]	Measures\Sales Measures	"â,¬"\ #,0.00;-"â,¬"\ #,0.00;"â,¬"\ #,0.00
Profit Margin		DIVIDE([Total Sales] - [Total Cost], [Total Sales])	Measures\Sales Measures	0.00%;-0.00%;0.00%
AOV	Average Order Value or Average Purchase Value	DIVIDE([Total Sales], DISTINCTCOUNT(fact_sales[order_id]))	Measures\Sales Measures	"â,¬"\ #,0.00;-"â,¬"\ #,0.00;"â,¬"\ #,0.00
Total Sales (PrevYear)		CALCULATE(SUMX(fact_sales, fact_sales[sub_total]),DATEADD(dim_date[date].[Date],-1,YEAR))	Measures\Sales Measures	
YoY Growth (%)		DIVIDE(SUMX(fact_sales,fact_sales[sub_total])-[Total Sales (PrevYear)],[Total Sales (PrevYear)])	Measures\Sales Measures	0.00%;-0.00%;0.00%
Last Date		MAXX(ALL(fact_sales), max(fact_sales[order_date].[Date]))	Measures\Date Measures	Short Date

RPR (%)	Repeat Purchase Rate	DIVIDE(CALCULATE(DISTINCTCOUNT(fact_sales[customer_id]), FILTER(VALUES(fact_sales[customer_id]), CALCULATE(COUNTROWS(fact_sales)) > 1)), CALCULATE(DISTINCTCOUNT(fact_sales[customer_id])))	Measures\Customer Measures	0.0%;-0.0%;0.0%
Total Orders		DISTINCTCOUNT(fact_sales[order_id])	Measures\Sales Measures	0
Total Customers (Prev Month)		CALCULATE([Total Customers], DATEADD(dim_date[date].[Date],-1,MONTH))	Measures\Customer Measures	0
Total Customers (Prev Year)		CALCULATE(COUNTX(fact_sales, fact_sales[customer_id]),DATEADD(dim_date[date].[Date], -1,YEAR))	Measures\Customer Measures	0
Total Customers		DISTINCTCOUNT(fact_sales[customer_id])	Measures\Customer Measures	0
APF	Average Purchase Frequency (APF)	DIVIDE([Total Orders],[Total Customers])	Measures\Customer Measures	
Total Sales (PrevMonth)		CALCULATE(SUMX(fact_sales, fact_sales[sub_total]),DATEADD(dim_date[date].[Date],-1,MONTH))	Measures\Sales Measures	
ARPU	Average Sales per Customer	DIVIDE([Total Sales], [Total Customers])	Measures\Sales Measures	
Highest Order Value by Sales Rep		CALCULATE(MAX(Fact_Sales[sub_total]), ALLEXCEPT(Fact_Sales, Fact_Sales[sales_rep_id]))	Measures\Sales Measures	
Avg Sales by Rep		AVERAGEX(VALUES(Fact_Sales[sales_rep_id]), CALCULATE(SUM(fact_sales[sub_total])))	Measures\Sales Measures	